G-MFRM: A Unified Diagnostic Framework for Joint Estimation of Judge Bias and Benchmark Contamination in LLM Evaluation

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Abstract

LLM-as-a-judge evaluation suffers from two confounded sources of distortion: judge severity bias and benchmark contamination. Existing methods address these independently, but sequential correction produces biased ability estimates due to omitted variable confounding—when only one source is modeled, the other's effect is absorbed into parameter estimates. We propose the Gated Many-Facet Rasch Model (G-MFRM), a staged joint calibration framework that simultaneously estimates model ability (θ_n), item difficulty (b_i), judge severity (d_j), and contamination susceptibility (c_n) via externally-anchored exposure indicators (e_i). On MT-Bench (160 items) and Arena-Hard (500 items) evaluated by 8 LLM judges across 22 models, G-MFRM achieves Kendall-τ = 0.823 with Chatbot Arena human rankings, outperforming all baselines including the best sequential method (UDA, τ = 0.805). The advantage concentrates on high-contamination models (top quartile: $\Delta\tau$ = +0.085, p = 0.021). Beyond ranking improvement, G-MFRM uniquely provides interpretable per-model contamination profiles (c_n correlates r = 0.52 with external ConStat validation), per-judge severity calibration (range: 1.74 logits), and per-item exposure diagnostics—a unified measurement framework no existing method offers.

1 Introduction

The use of large language models as evaluators (LLM-as-a-judge) has become the dominant paradigm for scalable assessment of language model quality (Zheng et al., 2023). However, this paradigm is compromised by two systematic distortions. First, LLM judges exhibit systematic severity differences—a phenomenon documented as inter-judge disagreement (Fleiss $\kappa \approx 0.38$) where different judges produce systematically different score distributions for identical model outputs (Zheng et al., 2023; Li et al., 2024). Second, benchmark contamination inflates model scores when evaluation items appear in training data, producing artificially high ability estimates for contaminated models (Golchin and Surdeanu, 2023; Shi et al., 2023).

These two problems have been addressed in parallel by separate research communities. Judge bias correction methods—CyclicJudge (Li et al., 2024), UDA (Son et al., 2024)—calibrate for inter-judge variation but ignore contamination. Contamination detection methods—ConStat (Oren et al., 2024), DCR (Dekoninck et al., 2024)—quantify data leakage but treat all judges as interchangeable. Sequential application of both corrections (in either order) fails to resolve the fundamental confounding between them.

The confounding problem. Judge severity and contamination susceptibility interact through shared items. A lenient judge evaluating a contaminated model on an exposed item produces an inflated score where both bias sources contribute additively—but their individual effects cannot be disentangled without simultaneous modeling. Formally, this is an instance of omitted variable bias (OVB) (Angrist and Pischke, 2009): when the covariance between judge severity and the contamination–exposure product is non-zero, sequential estimators are asymptotically biased regardless of estimation order.

Contributions. We propose G-MFRM, a staged joint calibration framework that:

1. **Identifies confounding**: We formalize the OVB structure in LLM evaluation and demonstrate via Monte Carlo simulation that sequential pipelines produce θ-RMSE 31.6% higher than joint estimation at realistic confounding levels ($\rho = 0.30$).

2. **Provides unified diagnostics**: G-MFRM produces interpretable per-model contamination susceptibility (c_n), per-judge severity (d_j), and per-item exposure (e_i) estimates—a four-dimensional diagnostic output no existing method provides.

3. **Achieves comparable or better rankings**: On combined MT-Bench and Arena-Hard data with 8 LLM judges evaluating 22 models, G-MFRM fair scores yield $\tau = 0.823$ with Chatbot Arena, outperforming UDA ($\tau = 0.805$) with the advantage concentrated on high-contamination models ($\Delta\tau = 0.085$, $p = 0.021$).

2 Motivating Observation: Confounding in LLM Evaluation

We begin with a formal characterization of why sequential correction fails.

2.1 Setup

Consider the data-generating process for LLM evaluation with N models, I items, and J judges. The probability that model n produces a response preferred by judge j on item i follows:

$$\log \frac{P(X_{nij} = 1)}{P(X_{nij} = 0)} = \theta_n - b_i - d_j + c_n \cdot e_i \tag{1}$$

where θ_n is model ability, b_i is item difficulty, d_j is judge severity (positive = harsher), and $c_n \cdot e_i$ captures the contamination inflation for model n on item i with exposure level $e_i \in [0, 1]$.

2.2 Omitted Variable Bias

Observation (OVB in LLM evaluation). *If Cov$(d_j, c_n \cdot e_i) \neq 0$ and only one of $\{d_j, c_n \cdot e_i\}$ is modeled, the omitted variable's effect is absorbed into remaining parameter estimates. The first-order bias on $\hat{\theta}_n$ is:*

$$\beta_{\text{OVB}} \approx \frac{\text{Cov}(d_j, c_n \cdot e_i)}{\text{Var}(d_j)} \tag{2}$$

This follows directly from the OVB formula of Angrist and Pischke (2009) applied to the multi-facet evaluation context. The condition $\text{Cov}(d_j, c_n \cdot e_i) \neq 0$ holds whenever harsher judges and contamination effects are non-orthogonal—which is the generic case in practice, since both relate to item difficulty.

2.3 Monte Carlo Validation

We validate this observation via Monte Carlo simulation (1,000 replications, $\rho = 0.30$). Table 1 reports the results.

Sequential estimators retain 0.11–0.13 logits of systematic bias regardless of estimation order, while joint G-MFRM reduces bias to 0.02 (a 31.6% RMSE reduction). A ρ-sweep (Table 2) confirms that sequential bias grows linearly with confounding strength while joint estimation remains approximately unbiased across all tested ρ values.

Table 1: Monte Carlo bias validation (1000 replications, $\rho = 0.30$). Joint G-MFRM achieves near-zero bias; sequential estimators retain systematic overestimation.

Estimator	Mean θ bias	θ RMSE
Joint G-MFRM	0.02	0.13
Sequential (bias-first)	0.13	0.19
Sequential (contam-first)	0.11	0.18
MFRM-only (ignore c)	0.14	0.20
Raw average	0.31	0.38

Table 2: Sequential estimator bias as a function of confounding strength $\rho(d_j, c_n \cdot e_i)$.

ρ	Sequential Bias	Joint Bias	Ratio
0.00	0.01	0.01	1.0
0.15	0.05	0.01	5.0
0.30	0.10	0.02	5.0
0.45	0.15	0.02	7.5
0.60	0.21	0.03	7.0

3 Method: Gated Many-Facet Rasch Model

G-MFRM operates in three stages to ensure identifiability while achieving conditional joint estimation.

3.1 Stage 1: Standard MFRM

We first fit a standard Many-Facet Rasch Model (Linacre, 1989; Eckes, 2011):

$$\log \frac{P(X_{nij} = 1)}{P(X_{nij} = 0)} = \theta_n - b_i - d_j \tag{3}$$

yielding initial estimates $\hat{\theta}_n^{(1)}$, $\hat{b}_i$, $\hat{d}_j$ via Bayesian MCMC. This stage establishes baseline ability estimates without contamination correction.

3.2 Stage 2: External Exposure Estimation

Per-item exposure $e_i \in [0, 1]$ is estimated via a three-source weighted composite:

$$\hat{e}_i = w_1 \cdot \text{Gap}_i + w_2 \cdot \text{Temporal}_i + w_3 \cdot \text{LogProb}_i \tag{4}$$

where Gap_i denotes the normalized performance gap between original and semantically-equivalent reference variants, Temporal_i captures pre/post-release score differentials, and LogProb_i measures verbatim memorization via log-probability analysis. Optimal weights ($w_1 = 0.62$, $w_2 = 0.18$, $w_3 = 0.20$) are determined by cross-validation against ConStat per-model deltas.

Stage 2 produces fixed $\hat{e}_i$ values that are treated as known in Stage 3. This staged approach is a deliberate identifiability choice: simultaneous estimation of all (θ, b, d, c, e) parameters is under-identified without external anchoring, analogous to the two-step estimator tradition in econometrics (Heckman, 1979).

Table 3: Judge coverage matrix (fraction of completed evaluations).

Judge	MT-Bench	Arena-Hard	Overall
GPT-5.5	0.99	0.98	0.984
Claude Opus 4.7	0.98	0.97	0.974
Claude Sonnet 4.6	0.99	0.98	0.984
DeepSeek-V4-Pro	0.97	0.95	0.958
Qwen3-235B-A22B	0.96	0.93	0.942
Qwen3.6-27B	0.94	0.91	0.922
Llama-4-Maverick	0.95	0.92	0.932
Gemini-2.5-Pro	0.98	0.96	0.968

3.3 Stage 3: Conditional Joint Estimation

With $\hat{e}_i$ fixed, we fit the full gated model:

$$\log \frac{P(X_{nij} = 1)}{P(X_{nij} = 0)} = \theta_n - b_i - d_j + c_n \cdot \hat{e}_i \tag{5}$$

where $c_n \geq 0$ (Half-Cauchy prior) represents model n's contamination susceptibility. The gating mechanism $c_n \cdot \hat{e}_i$ activates proportionally to item exposure—when $\hat{e}_i \approx 0$, the model reduces to standard MFRM, ensuring graceful degradation.

Identifiability. With $\hat{e}_i$ fixed, c_n is identified from differential performance patterns across items with varying exposure levels. We require ≥ 4 judges for stable separation of d_j and c_n effects.

Inference. We use Hamiltonian Monte Carlo with 4 chains, 3000 warmup iterations, 5000 sampling iterations with thinning factor 2.

4 Experimental Setup

4.1 Data Collection

We evaluate 22 target models across two benchmarks: MT-Bench (160 items across 8 categories) and Arena-Hard (500 items), using 8 diverse LLM judges (Table 3). Each evaluation includes position-swapped pairwise comparisons, yielding 219,874 total judgments (94.6% collection completion rate, 3.1% API failure, 2.3% refusal).

Inter-judge Fleiss $\kappa = 0.38$ confirms substantial systematic disagreement. Same-family judge pairs show higher agreement (e.g., Claude Opus–Sonnet: $\kappa = 0.51$) than cross-family pairs (mean $\kappa = 0.32$), consistent with preference leakage effects.

4.2 Per-Item Exposure Estimation

We construct reference variants for 627 items (87% quality pass rate). The performance gap source achieves 95% coverage, while temporal signal (41%) and LogProber (55%) provide complementary but sparser signals. The composite $\hat{e}_i$ distribution has mean 0.34 (std 0.22), with MT-Bench items showing higher exposure (mean 0.42) than Arena-Hard (mean 0.29), consistent with MT-Bench's earlier release and broader training data inclusion.

Table 4: Per-source e_i estimation statistics.

Source	Coverage	Mean e_i	Std	Weight
Performance Gap	0.95	0.36	0.24	0.62
Temporal Signal	0.41	0.31	0.19	0.18
LogProber	0.55	0.33	0.26	0.20
Composite	1.00	0.34	0.22	—

Table 5: Method comparison on MT-Bench + Arena-Hard (Chatbot Arena Elo as reference ranking).

Method	τ	ρ	Top-5	Top-10	$\lvert\Delta R\rvert$
Raw Average	.761	.812	0.6	0.7	3.1
MFRM-only	.794	.831	0.8	0.8	2.1
Seq. (bias-first)	.788	.824	0.6	0.8	2.4
Seq. (contam-first)	.791	.828	0.8	0.8	2.2
CyclicJudge	.802	.838	0.8	0.8	1.9
CyclicJudge + Disc.	.798	.835	0.8	0.8	2.0
DCR Factor	.779	.821	0.6	0.7	2.6
UDA	.805	.841	0.8	0.8	1.8
PrefLeak Calib.	.783	.822	0.6	0.8	2.5
G-MFRM (ours)	**.823**	**.857**	**0.8**	**0.9**	**1.5**

4.3 Baselines

We compare G-MFRM against: (1) **Raw Multi-Judge Average**, (2) **MFRM-only** (Eq. 3), (3–4) **Sequential pipelines** (bias-first and contamination-first), (5) **CyclicJudge** (Li et al., 2024), (6) **CyclicJudge + Discard** (removing high-exposure items), (7) **DCR Factor** (Dekoninck et al., 2024), (8) **UDA** (Son et al., 2024), and (9) **PrefLeak Calibration**. Chatbot Arena Elo serves as the reference human ranking.

5 Results

5.1 Main Comparison

Table 5 presents the primary results. G-MFRM achieves the highest Kendall-τ (0.823) and Spearman-ρ (0.857) with Chatbot Arena rankings, outperforming all baselines.

The improvement over UDA ($\Delta\tau = +0.018$, 95% CI [0.002, 0.034]) is borderline significant ($p = 0.067$, Wilcoxon signed-rank). G-MFRM significantly outperforms MFRM-only ($p = 0.042$), sequential pipelines ($p \leq 0.024$), DCR ($p = 0.008$), and PrefLeak ($p = 0.011$).

CyclicJudge + Discard performs *worse* than CyclicJudge alone (τ = 0.798 vs 0.802), confirming that item discard wastes information and increases estimation variance at small sample sizes.

5.2 Subset Analysis: High-Contamination Models

The key finding emerges from stratifying by contamination level (Table 6). G-MFRM's advantage is concentrated on models in the top contamination quartile ($c_n > 0.75$).

For low-contamination models (Q1), G-MFRM gracefully degrades to MFRM-level performance (no overcorrection). For high-contamination models, the improvement is substantial and statistically significant ($\Delta\tau = +0.085$, $p = 0.021$), with post-hoc power of 0.74.

Table 6: G-MFRM advantage by model contamination quartile (vs. UDA).

Quartile	N	c_n Range	$\tau_{\text{G-MFRM}}$	τ_{UDA}	$\Delta\tau$
Q1 (lowest)	6	[0.04, 0.18]	.807	.811	$-.004$
Q2	5	[0.22, 0.48]	.815	.802	+.013
Q3	5	[0.52, 0.75]	.831	.808	+.023
Q4 (highest)	6	[0.85, 1.42]	**.867**	.782	**+.085***

$^*p = 0.021$ (Wilcoxon signed-rank).

Table 7: Estimated judge severity d_j (positive = harsher).

Judge	d_j (mean)	95% CI
GPT-5.5	-0.83	$[-1.02, -0.64]$
Claude Opus 4.7	-0.31	$[-0.48, -0.14]$
Claude Sonnet 4.6	-0.12	$[-0.29, +0.05]$
Gemini-2.5-Pro	-0.22	$[-0.40, -0.04]$
Llama-4-Maverick	+0.15	$[-0.03, +0.33]$
Qwen3-235B-A22B	+0.44	[+0.26, +0.62]
Qwen3.6-27B	+0.62	[+0.43, +0.81]
DeepSeek-V4-Pro	+0.91	[+0.72, +1.10]

5.3 Diagnostic Outputs

Judge severity profiles. Table 7 presents estimated d_j values. The severity range spans 1.74 logits (GPT-5.5: $d = -0.83$ to DeepSeek-V4-Pro: $d = +0.91$), corresponding to approximately 1.5 rating scale levels of systematic bias. All judges show acceptable model fit (Infit MNSQ: [0.94, 1.21]).

Per-model contamination susceptibility. The fitted c_n distribution (median 0.48, range [0.04, 1.42]) correlates with external ConStat per-model deltas at $r = 0.52$ ($p = 0.013$), supporting construct validity. Models with highest c_n (Qwen3-235B-A22B: 1.42, Qwen3.6-27B: 0.85, GPT-4.5: 0.72) show the largest ability estimate corrections in Stage 3.

Theta overestimation by contamination level. The mean $\hat{\theta}$ overestimation (Stage 1 vs Stage 3) scales with c_n: low-c_n models ($c_n < 0.25$, $N = 6$) show negligible bias (mean 0.03 logits), medium-c_n models (0.25–0.75, $N = 10$) show moderate bias (0.11 logits), and high-c_n models ($c_n > 0.75$, $N = 6$) show substantial bias (0.21 logits). This directly confirms the confounding mechanism identified in Section 2.

5.4 Model Fit and Convergence

Stage 1 achieves $\hat{R}_{\max} = 1.02$ and $\text{ESS}_{\min} = 412$. Stage 3 shows acceptable but weaker convergence ($\hat{R}_{\max} = 1.06$, $\text{ESS}_{\min} = 132$) with 14 divergent transitions (0.07% of samples). WAIC model comparison favors G-MFRM over MFRM (ΔWAIC = -489 ± 124, approximately 4σ significant), providing statistical evidence for the contamination term beyond ranking metrics.

Person-fit statistics (l_z) correlate moderately with c_n ($r = 0.41$, $p = 0.058$), indicating that traditional misfit detection captures part of but not all contamination signal—supporting c_n as a distinct and complementary parameter.

Table 8: Per-model c_n estimates (top-10 by Arena Elo) with ConStat validation. θ shift: Stage 1 $\rightarrow$ Stage 3.

Model	c_n	95% CI	ConStat Δ	θ shift
GPT-5.5	0.31	[0.08, 0.58]	0.21	-0.04
Claude Opus 4.7	0.22	[0.03, 0.47]	0.15	-0.02
Gemini-2.5-Pro	0.58	[0.29, 0.91]	0.44	-0.12
DeepSeek-V4-Pro	0.15	[0.01, 0.38]	0.11	+0.01
Claude Sonnet 4.6	0.41	[0.15, 0.71]	0.28	-0.08
Llama-4-Maverick	0.09	[0.00, 0.28]	0.07	+0.03
Qwen3-235B-A22B	1.42	[0.98, 1.91]	0.82	-0.27
GPT-4.5	0.72	[0.41, 1.08]	0.51	-0.15
Mistral-Large-3	0.38	[0.12, 0.68]	0.30	-0.05
Qwen3.6-27B	0.85	[0.52, 1.22]	0.61	-0.19

Table 9: Facet fit statistics summary.

Facet	N	Infit Range	Outfit Range	Misfit %
Model (θ)	22	[0.82, 1.28]	[0.78, 1.35]	9.1%
Item (b)	660	[0.71, 1.42]	[0.65, 1.58]	5.8%
Judge (d)	8	[0.94, 1.21]	[0.88, 1.31]	12.5%

6 Analysis

6.1 Exposure Indicator Ablation

Table 10 presents leave-one-out ablation of the e_i estimation sources. The performance gap source contributes most ($\Delta\tau = -0.037$ when removed), while temporal signal contributes least ($\Delta\tau = -0.005$). A performance-gap-only configuration achieves $\tau = 0.814$—only 0.009 below the full pipeline—providing a simplified deployment option.

6.2 Judge Count Sensitivity

Table 11 shows G-MFRM performance as a function of the number of judges. The model requires ≥ 4 judges to reliably separate d_j from c_n effects. With 2 judges, G-MFRM gracefully degrades to MFRM-level performance ($\Delta\tau = +0.003$, $\hat{R} = 1.18$). We recommend 6+ judges for practical deployment.

Preference leakage diagnostic. Leave-one-judge-out analysis detects 2 significant same-family preference leakage pairs: Claude Opus 4.7 $\rightarrow$ Claude Sonnet 4.6 (residual shift -0.14, $p = 0.032$) and Qwen3-235B-A22B $\rightarrow$ Qwen3.6-27B (residual shift -0.11, $p = 0.048$). These effects are small relative to the d_j range (1.74 logits) and are absorbed uniformly into $\hat{\theta}_n$.

6.3 Cross-Benchmark Generalization

We replicate the G-MFRM pipeline on WildBench (1,024 items, 6 judges, 18 models) as a held-out generalization test (Table 12).

Per-model c_n estimates show high cross-benchmark consistency: $r = 0.78$ (MT-Bench vs Arena-Hard), $r = 0.74$ (Arena-Hard vs WildBench), $r = 0.68$ (MT-Bench vs WildBench). This confirms contamination susceptibility is a model-level trait, not a benchmark artifact.

Table 10: e_i source ablation: impact on downstream G-MFRM performance.

Configuration	N Sources	τ	$\Delta\tau$
Full Pipeline	3	.823	—
w/o Performance Gap	2	.786	$-.037$
w/o Temporal Signal	2	.818	$-.005$
w/o LogProber	2	.812	$-.011$
Only Performance Gap	1	.814	$-.009$
Only Temporal	1	.788	$-.035$
Only LogProber	1	.795	$-.028$
MFRM (no e_i)	0	.794	$-.029$

Table 11: G-MFRM performance by number of judges.

J	$\tau_{\text{G-MFRM}}$	τ_{MFRM}	Δ	$\hat{R}_{\max}$	$\text{ESS}_{\min}$
2	.768	.765	+.003	1.18	41
4	.791	.779	+.012	1.09	84
6	.812	.789	+.023	1.07	108
8	.823	.794	+.029	1.06	132

6.4 Error Propagation Sensitivity

Since Stage 2 $\hat{e}_i$ values carry estimation noise, we assess robustness by injecting Gaussian noise at varying σ levels (Table 13).

At $\sigma \leq 0.2$ (our estimated operating range), θ RMSE degrades by only 6.3% and c_n correlation with truth remains above 0.84. The staged estimation is robust within practical noise levels. At $\sigma = 0.5$ (worst case), degradation becomes substantial (36.7% θ RMSE increase), establishing a clear boundary condition.

6.5 Adversarial Robustness

We test G-MFRM against three adversarial contamination types following Wang et al. (2026) (Table 14).

G-MFRM provides a rank protection rate of 66%, reducing mean rank corruption from 3.2 to 1.1 positions. Detection performance scales with severity (AUC: 0.78 mild $\rightarrow$ 0.91 severe). The fundamental limitation is adversarial contamination that mimics genuine ability gain uniformly across items (22% undetectable)—this is outside the rank-1 $c \cdot e$ gating assumption.

6.6 Post-Hoc Power Analysis

With $N = 22$ models, the study achieves post-hoc power of 0.58 for detecting the observed effect size ($d = 0.42$) at $\alpha = 0.05$. Power reaches 0.74 when restricted to the high-contamination subset ($N = 6$, $d = 1.31$). An estimated $N = 35$ models would be required for 80% power on the full sample, indicating that our study is moderately underpowered for the overall comparison but adequately powered for the clinically relevant subset.

7 Conclusion

We presented G-MFRM, a staged joint calibration framework that addresses the previously unrecognized confounding between judge bias and benchmark contamination in LLM evaluation. Our formal analysis

Table 12: Cross-benchmark performance and c_n consistency.

Benchmark	Items	J	$\tau_{\text{G-MFRM}}$	τ_{UDA}	$\Delta\tau$	c_n med.
MT-Bench	160	8	.811	.798	+.013	0.55
Arena-Hard	500	8	.831	.812	+.019	0.44
WildBench	1024	6	.808	.793	+.015	0.38
Combined	660	8	.823	.805	+.018	0.48

Table 13: Error propagation sensitivity (20 replications per σ).

σ_e	θ RMSE	θ τ	c_n RMSE	c_n Corr
0.0	0.128	.916	0.53	0.86
0.1	0.130	.915	0.55	0.85
0.2	0.136	.914	0.58	0.84
0.5	0.175	.885	0.84	0.75

establishes that sequential correction pipelines retain systematic bias proportional to confounding strength, while G-MFRM's conditional joint estimation achieves near-zero bias.

On real evaluation data (22 models, 8 judges, 660 items), G-MFRM achieves the highest agreement with human rankings ($\tau = 0.823$) with particular strength on high-contamination models ($\Delta\tau = +0.085$ in top quartile). Beyond ranking accuracy, G-MFRM provides interpretable diagnostic outputs—per-judge severity profiles (1.74 logit range), per-model contamination susceptibility (validated at $r = 0.52$ against ConStat), and per-item exposure indicators—constituting a unified measurement framework for LLM evaluation auditing that no existing method offers.

Limitations

Statistical power. With $N = 22$ models, the overall comparison against UDA achieves only 0.58 post-hoc power. The borderline significance ($p = 0.067$) means we cannot claim definitive ranking superiority over UDA on the full population. Future work should expand to ≥ 35 models.

Exposure estimation noise. The effective $\sigma_e \approx 0.24$ of our e_i pipeline sits at the edge of the robust regime ($\sigma \leq 0.2$), introducing 6–10% θ precision loss relative to oracle exposure values. Temporal signal coverage (41%) limits the composite quality.

Rank-1 assumption. The $c_n \cdot e_i$ gating assumes contamination effects follow a rank-1 structure. Adversarial contamination that uniformly inflates performance across all items (22% of tested adversarial patterns) is fundamentally undetectable under this assumption.

Judge-model interaction. G-MFRM does not model f_{nj} (judge-model interaction). Same-family preference leakage (detected in 2/5 pairs, effect ≤ 0.14 logits) is absorbed uniformly into $\hat{\theta}_n$ rather than explicitly corrected.

Convergence. Stage 3 convergence ($\hat{R} = 1.06$, ESS = 132) is acceptable but not ideal. The 14 divergent transitions (0.07%) and marginal posterior coverage (c_n 95% CI coverage: 88% vs nominal 95%) suggest the Half-Cauchy prior may be slightly too informative for high-c_n models.

Table 14: Adversarial contamination detection and rank protection.

Type	Severity	AUC	Rank Corrupt.	
			w/o	w/ G-MFRM
Data Poisoning	Mild	0.76	1.8	0.9
Data Poisoning	Severe	0.92	4.1	1.2
Prompt Injection	Mild	0.74	1.5	0.8
Prompt Injection	Severe	0.88	3.8	1.4
Memorization	Mild	0.83	2.1	0.7
Memorization	Severe	0.93	4.5	0.9

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